Answer Key & Marking Scheme

Subject: Science | **Test:** Test 3 (Day 21, 30 May 2026) — Weekly
Maximum Marks: 80

Syllabus: How do Organisms Reproduce? • Heredity • Light – Reflection and Refraction

SECTION A — Answer Key (1 mark each)

Q1. (b) Yeast.

Q2. (c) Rhizopus.

Q3. (c) fallopian tube.

Q4. (c) egg cell.

Q5. (d) Insulin.

Q6. (b) gene.

Q7. (d) all of these.

Q8. (d) any of A, B, AB or O — depends on whether the parents are heterozygous.

Q9. (b) the father — the X or Y chromosome from the father determines the sex.

Q10. (b) a virtual, erect, magnified image — when the face is placed within the focal length.

Q11. (b) infinity.

Q12. (b) virtual, erect, diminished.

Q13. (b) absolute refractive index.

Q14. (d) both (b) and (c).

Q15. (b) –5 D — P = 1/f (in m) = 1/(–0.20) = –5 D.

Q16. (b) between F and 2F.

Q17. Homozygous: an individual having two identical alleles for a particular gene (e.g. TT or tt).

Q18. Dioptre (D), where 1 D = 1 m■^{1}.

Q19. Pollination is the transfer of pollen grains from the anther of a stamen to the stigma of a carpel of a flower.

Q20. (i) The angle of incidence equals the angle of reflection. (ii) The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.

SECTION B — Answer Key (2 marks each)

Q21. Sexual reproduction involves two parents, formation of gametes and fertilisation, leading to genetic variation. Asexual reproduction involves a single parent and the offspring is genetically identical to the parent. Sexual is slower; asexual is faster. (2)

Q22. Vegetative propagation is the formation of new plants from vegetative parts (root, stem, leaf) without seeds. Examples: potato (stem tuber), bryophyllum (leaf), rose (stem cutting), ginger (rhizome). (2)

Q23. The placenta is a special tissue that connects the developing embryo to the mother's uterine wall. It allows: exchange of nutrients and oxygen from mother to embryo, and removal of waste from embryo to mother's blood. (2)

Q24. Variations help organisms survive in changing environmental conditions. If a population is genetically identical, a single environmental change (e.g. disease, heat) could wipe it out. Variations ensure that some individuals survive and pass on their genes — supporting evolution. (2)

Q25. Using mirror formula: 1/f = 1/v + 1/u. With sign convention: u = –60 cm, v = –30 cm. 1/f = 1/(–30) + 1/(–60) = (–2 – 1)/60 = –3/60 = –1/20. So **f = –20 cm**. (2)

Q26. Snell's law: When a ray of light passes from one medium to another, sin i / sin r = constant = n■■ (refractive index of medium 2 relative to medium 1). The incident ray, refracted ray and normal all lie in the same plane. (2)

SECTION C — Answer Key (3 marks each)

Q27. After pollination, the pollen grain lands on the stigma. It germinates and grows a pollen tube down the style. The tube carries two male gametes to the ovule. One male gamete fuses with the egg cell to form a zygote (fertilisation), and the other fuses with the polar nuclei to form the endosperm — this is called **double fertilisation**, characteristic of angiosperms. (3)

Q28. (i) **Ovaries** — produce eggs (ova) and female sex hormones. (ii) **Fallopian tubes** — site of fertilisation; carry the egg from ovary to uterus. (iii) **Uterus** — where the embryo develops; site of implantation. (iv) **Vagina** — birth canal. (3)

Q29. Mendel crossed pure tall (TT) with pure dwarf (tt). **F■**: all Tt — tall. On self-pollination of F■: **F■** — TT : Tt : tt = 1 : 2 : 1; phenotype tall : dwarf = 3 : 1. This shows that traits are inherited as discrete units; recessive traits reappear in F■. This is Mendel's **Law of Segregation**. (3)

Q30. u = –25 cm, f = –15 cm. 1/v = 1/f – 1/u = 1/(–15) – 1/(–25) = –1/15 + 1/25 = (–5 + 3)/75 = –2/75. v = –75/2 = –37.5 cm. (Real image, on same side as object.) Magnification m = –v/u = –(–37.5)/(–25) = –1.5. Image height = 1.5 × 5 = –7.5 cm (inverted, magnified). **Position: 37.5 cm in front of mirror; size: 7.5 cm; nature: real, inverted, magnified.** (3)

Q31. Total internal reflection: when a ray of light travels from an optically denser medium to a rarer medium and the angle of incidence exceeds the critical angle, the ray is reflected back entirely into the denser medium. **Conditions:** (i) light must travel from denser to rarer medium; (ii) angle of incidence must be greater than the critical angle. (3)

Q32. Centre of curvature (C): centre of the sphere of which the spherical mirror is a part. **Principal axis:** straight line passing through the centre of curvature and the pole of the mirror. **Pole (P):** the geometrical centre of the reflecting surface of the mirror. (3)

Q33. u = –20, f = +10. Lens formula: 1/v – 1/u = 1/f ⇒ 1/v = 1/10 + 1/(–20) = 2/20 – 1/20 = 1/20. v = +20 cm. m = v/u = 20/(–20) = –1. So image is at 20 cm on the other side, real, inverted, of the same size as the object. (3)

SECTION D — Answer Key (5 marks each)

Q34. (a) Diagram with labels: sepal, petal, stamen (anther + filament), carpel/pistil (stigma + style + ovary + ovule). (3) (b) After fertilisation: the zygote develops into the embryo; the ovule becomes the seed; the ovary develops into the fruit. The other floral parts (sepals, petals, stamens) usually wither and fall off. (2)

Q35. (a) *Laws of refraction:* (i) The incident ray, the refracted ray and the normal at the point of incidence lie in the same plane. (ii) Sin i / Sin r = n■■ (constant) for a given pair of media (Snell's law). (2) (b) The refractive index of a medium is the ratio of the speed of light in vacuum to the speed of light in the medium: n = c/v. (1) (c) n = c/v = 3·10■ / 2·10■ = **1.5**. (2)

Q36. (a) *Law of Segregation:* During gamete formation, the two alleles of a gene segregate so that each gamete receives only one allele. Example: tall × dwarf cross — F■ all tall, F■ in 3:1 ratio. *Law of Independent Assortment:* alleles of different genes segregate independently. Example: dihybrid cross with round-yellow and wrinkled-green peas gave 9:3:3:1 ratio in F■. (3) (b) Females have XX and males have XY chromosomes. The mother always passes an X. The father may pass X (giving daughter, XX) or Y (giving son, XY). Punnett square: father X & Y crossed with mother X & X → XX, XY (each with 50% probability). Hence the sex of the child is determined by the chromosome contributed by the father. (2)

SECTION E — Answer Key (4 marks each)

Q37. (i) Any two of: condom, IUCD (Cu-T), oral contraceptive pills, tubectomy/vasectomy. (1) (ii) Cu-T causes a mild inflammatory reaction in the uterus that prevents implantation of the fertilised egg. (1) (iii) Possible side-effects like hormonal imbalance, weight gain, mood changes, increased risk of certain conditions; not protective against STIs. (1) (iv) Barrier methods like condoms also protect against STIs (HIV/AIDS, syphilis, etc.) — other methods do not. (1)

Q38. (i) Ravi has attached earlobes; since this is recessive, his genotype must be **ee**. (1) (ii) Both parents are 'free' but had a child with 'attached'. So both parents must be heterozygous: **Ee × Ee**. (1) (iii) Each parent passes either E or e to the child with equal probability. With 25% probability the child gets ee from both — and shows the recessive trait (attached earlobes), even though both parents are phenotypically 'free'. (2)

Q39. (i) Within the focal length of the lens (between the lens and F). (1) (ii) Virtual, erect and magnified. (1) (iii) Ray diagram: object placed between lens and F; one ray parallel to principal axis refracts through F on opposite side; another ray through optical centre passes undeviated; the rays diverge after refraction — extended backward they appear to meet on the same side as the object, giving a virtual, erect, magnified image. (2)

— End of Answer Key —